13

Fourier Transform and image compression

In this chapter, we discuss the background of the Fourier Transform from electrical engineering and signal processing. Its fundamental property of transforming a (discrete or continuous) signal from its description in the time domain to an equivalent characterization in the frequency domain is used to describe and analyze the contributions of different frequencies to a signal. Furthermore, we present an application of the Fourier Transform in image processing.

13.1. The Continuous and the Discrete Fourier Transform

DEFINITION 13.1. *A **continuous signal** (or **analog signal**) is a function*

$$f: D \longrightarrow \mathbb{R}^n, \text{ where } D \subseteq \mathbb{R}^m \text{ and } m, n \in \mathbb{N}.$$

*A **discrete(-time) signal** is a function*

$$f: D \longrightarrow \mathbb{R}^n, \text{ where } D \subseteq \mathbb{Z}^m \text{ and } m, n \in \mathbb{N}.$$

*If in addition $f(D) \subseteq \mathbb{Z}^n$, then f is called a **digital signal**.*

Sound is a continuous signal that varies over time and has range loudness. It is an example of a signal $f: \mathbb{R} \longrightarrow \mathbb{R}$. In the case of gray-scale pixels on a screen, the signal associates to each point an intensity value, so that $f: D \subseteq \mathbb{Z}^2 \longrightarrow \mathbb{R}$. When color is represented by the constituent amounts of three basic colors (RGB) or four basic colors (CMYK), we have a signal mapping into $\mathbb{R}^3$ or $\mathbb{R}^4$, respectively.

A discrete signal is often obtained by sampling a continuous signal at discrete intervals. This is illustrated in Figure 13.1, where a continuous signal is sampled at regularly spaced points.

Discrete signals find applications in areas such as biomedical engineering, seismology, acoustics, sonar and radar imaging, speech communication, data communication, television satellite communication, satellite images, and many more. Speech and telephone signals are examples of signals with only one dimensional

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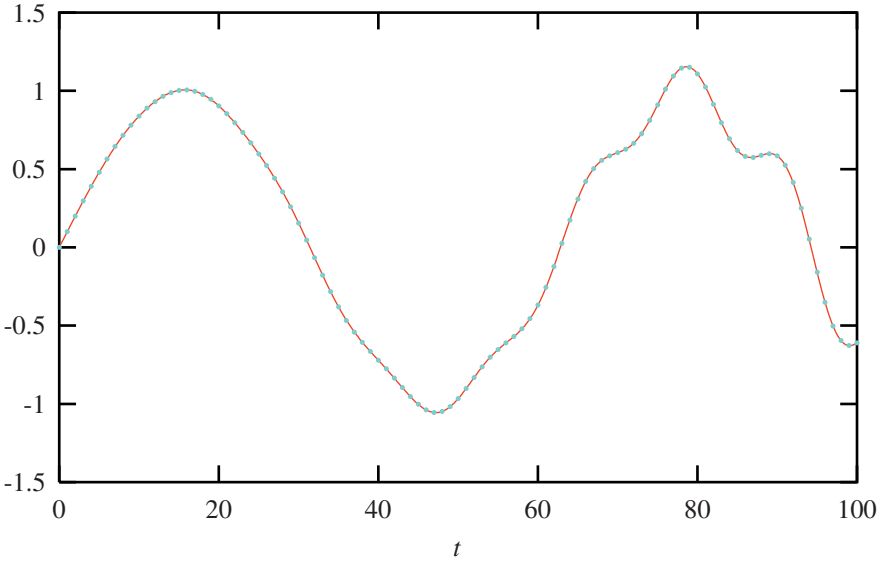


FIGURE 13.1: The analog signal $f(t) = \sin(t/10) + t^2 \sin(t/2)/40000$ (red curve), and the corresponding discrete signal (blue dots).

domains, while radar imaging, satellite images, and lunar images are processed with two dimensional domains. When modeling complicated problems, such as those that appear in seismology, the domain can have many dimensions.

It is important to perform certain operations on signals to extract relevant information from them, or to transform the signal to make it easier to use. For instance one may wish to extract some important parameters from the data such as danger alerts from an electrocardiogram or electroencephalogram. One may want to compress the data contained in a telephone signal or recognize the words associated with speech signals. A common problem is to extract relevant information from masses of data associated with such things as television transmission or satellite images. Another application of signal processing in signal transmission is to try to remove signal interference contributed by transmission noise, fading or channel distortion.

Of particular importance are the sine signal $f:\mathbb{R} \longrightarrow \mathbb{R}$ with $f(t) = \sin t$, and its complex variant $f:\mathbb{R} \longrightarrow \mathbb{R}^2 \cong \mathbb{C}$ with $f(t) = e^{it} = \cos t + i \sin t$, where $i = \sqrt{-1}$. More generally, we have the signal $f:\mathbb{R} \longrightarrow \mathbb{C}$ with $f(t) = a \cdot e^{ikt}$, with **amplitude** a, corresponding to the intensity of the signal (for example, loudness of an audio signal or luminance of a video signal), and **frequency** k (corresponding to pitch or color, respectively). All those signals are examples of **periodic** signals: there exists a **period** $T \in \mathbb{R}_{>0}$ such that $f(t+T) = f(t)$ for all $t \in \mathbb{R}$. Applying the transformation $t \longmapsto 2\pi t/T$, we may assume that $T = 2\pi$. For the sinusoidal signal $f(t) = a \cdot e^{ikt}$, the smallest such period T is the wavelength and related to

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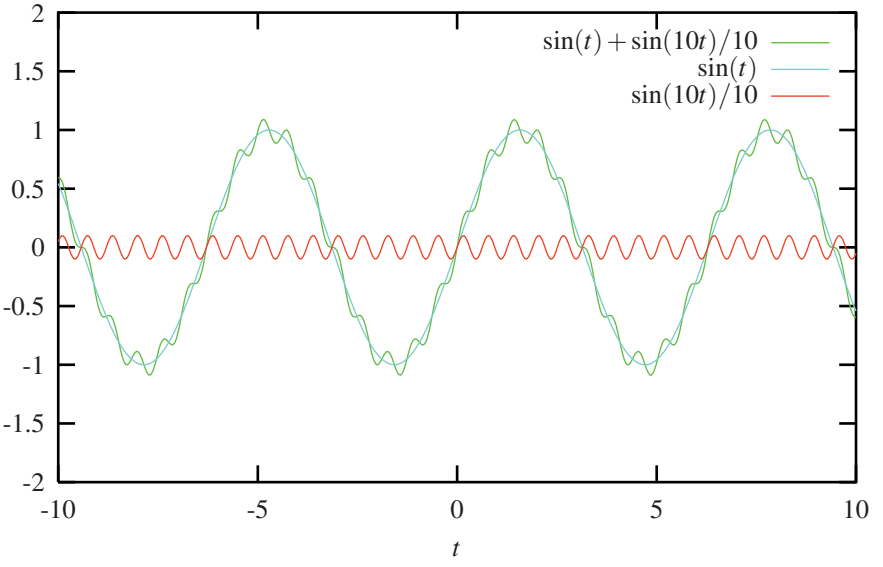


FIGURE 13.2: A 2π-periodic signal and its harmonics.

the frequency k by $T = 2\pi/k$. Similarly, a discrete signal is periodic with period $n \in \mathbb{N}_{>0}$ if $f(j+n) = f(j)$ for all $j \in \mathbb{Z}$.

Other popular examples of 2π-periodic functions are obtained by taking a function defined on a bounded interval, normalizing the interval to be $[0, 2\pi]$, and extending the function by periodicity. Examples are given in Exercise 13.4. In the following, we assume without further mention that our functions are sufficiently "smooth", so that, for example, the required integrals and sums are well defined.

DEFINITION 13.2. *The **(Continuous) Fourier Transform** of a 2π-periodic signal $f: \mathbb{R} \longrightarrow \mathbb{C}$ is $\widehat{f}: \mathbb{Z} \longrightarrow \mathbb{C}$ with*

$$\widehat{f}(k) = \int_0^{2\pi} f(t) e^{-ikt} dt \text{ for } k \in \mathbb{Z},$$

where $i = \sqrt{-1} \in \mathbb{C}$ as usual.

The following inversion formula expresses the function f in terms of its Fourier Transform:

$$f(t) = \frac{1}{2\pi} \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikt}. \tag{1}$$

This holds for all $t \in \mathbb{R}$; the series converges uniformly to f. This series is called the **Fourier series** of f, and the numbers $\beta_k = \frac{1}{2\pi}\widehat{f}(k) = \frac{1}{2\pi}\int_0^{2\pi} f(t) e^{-ikt} dt$ for

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$k \in \mathbb{Z}$ are the **Fourier coefficients** of f. The inversion formula says that the function f is uniquely determined by the sequence of its Fourier coefficients $(\beta_k)_{k\in\mathbb{Z}}$. The special functions e^{ikt}, for $k \in \mathbb{Z}$, are a "basis" for the complex vector space of all 2π-periodic functions; however, there are in general infinitely many nonzero coefficients.

While the original signal f is described in the time domain by assigning to each time $t \in [0, 2\pi]$ the value $f(t)$ of the signal at that time, the Fourier Transform $\widehat{f}$ is an equivalent characterization of f in the frequency domain. It associates to each frequency $k \in \mathbb{Z}$ the contribution $\widehat{f}(k)$ of that frequency, namely of the signal $\exp(ikt)/2\pi$, to f, as given by the inversion formula (1). It compresses the "continuously" many values $f(t)$ into the countably many values $\widehat{f}(k)$. For $k \in \mathbb{N}_{>0}$, the signal $(\widehat{f}(k)\exp(ikt) + \widehat{f}(-k)\exp(-ikt))/2\pi$ is called the kth **harmonic** of f.

EXAMPLE 13.3. Consider the 2π-periodic signal $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(t) = \sin(t) + \sin(10t)/10$, which is plotted in green in Figure 13.2. We may think of f as consisting of a low frequency part $\sin(t)$ with a large amplitude plus a high frequency part $\sin(10t)/10$ with a small amplitude; these are the blue and red curves in Figure 13.2. The Fourier Transform decomposes f into its harmonics: Exercise 13.3 shows that

$$\widehat{f}(1) = -\pi i = -\widehat{f}(-1), \quad \widehat{f}(10) = -\frac{1}{10}\pi i = -\widehat{f}(-10),$$

and $\widehat{f}(k) = 0$ for $k \neq \pm 1, \pm 10$. Thus the first harmonic of f is $(\widehat{f}(1)\exp(it) + \widehat{f}(-1)\exp(-it))/2\pi = \sin(t)$, the 10th harmonic is $\sin(10t)/10$, and all other harmonics are zero. $\diamond$

The **Discrete Fourier Transform** is the analog of the Continuous Fourier Transform for discrete periodic signals. If $f\colon \mathbb{Z} \longrightarrow \mathbb{C}$ is a discrete signal with period $n \in \mathbb{N}_{>0}$, its Discrete Fourier Transform $\widehat{f}\colon \mathbb{Z} \longrightarrow \mathbb{C}$ is defined by

$$\widehat{f}(k) = \sum_{0\le j<n} f(j)e^{-2\pi ijk/n} = \sum_{0\le j<n} f(j)\,\omega^{kj} \text{ for } k \in \mathbb{Z},$$

where $\omega = e^{-2\pi i/n} \in \mathbb{C}$ is a primitive nth root of unity. If we start with a continuous 2π-periodic signal g and sample it at regularly spaced points $2\pi j/n$ for $0 \le j < n$ to obtain f, then the Discrete Fourier Transform of f is just one of the common approximations to the integral defining the Continuous Fourier Transform of g. In contrast to the continuous case, $\widehat{f}$ is again periodic with period n. The analog of the inversion formula (1) is

$$f(j) = \frac{1}{n}\sum_{0\le k<n} \widehat{f}(k)e^{2\pi ijk/n} = \frac{1}{n}\sum_{0\le k<n} \widehat{f}(k)\,\omega^{-kj}. \tag{2}$$

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The connection to Section 8.2 is as follows. If we associate to a polynomial $g = \sum_{0 \le j < n} g_j x^j \in \mathbb{C}[x]$ a discrete signal $f: \mathbb{Z} \longrightarrow \mathbb{C}$ of period n with $f(j) = g_j$ for all j and take $\omega = e^{-2\pi i/n}$, then

$$\mathrm{DFT}_\omega(g) = \Big(\widehat{f}(0), \ldots, \widehat{f}(n-1)\Big),$$

and Theorem 8.13 is the analog of the inversion formula (2).

In digital signal processing, continuous signals such as sounds or images are sampled. The analysis or modification of the corresponding discrete signals (for example, noise reduction or contrast amplification) on digital hardware or software systems usually involves the computation of Discrete Fourier Transforms for large amounts of data, and the use of the Fast Fourier Transform is indispensable.

13.2. Audio and video compression

The Fourier Transform and its inverse can be considered as a change of representation between the time domain and the frequency domain. If f is a (continuous or discrete) signal and $\widehat{f}$ its Fourier Transform, then $f(t)$ is the value of the signal at time t, while $\widehat{f}(k)$ is the contribution of the frequency k to the signal. Both are equivalent characterizations of a signal. Often audio and video data intended for reception by humans tend to vary "slowly" in the time domain, so that their high frequency contributions are "small". In the frequency domain, this means that $|\widehat{f}(k)|$ is small for large values of k. The idea for data compression is now to discard the values of $\widehat{f}(k)$ that are "close to zero" and to only store the rest. Since the perceptive faculty of the human ear or eye is much better for low frequencies than for high ones, the listener or observer hardly notices this loss of information, and considerable compression rates can be achieved.

Audio and video data are real-valued signals, and often a variant of the Fourier Transform is used that maps real-valued signals again to real-valued signals, as follows. Let $f: \{0, \ldots, n-1\} \longrightarrow \mathbb{R}$ be a discrete real-valued signal of finite duration $n \in \mathbb{N}$, for example, a sampled piece of music or one row of a digital screen image. We may think of f as being extended to a periodic signal on $\mathbb{Z}$ by letting $f(j) = f(j \text{ rem } n)$ for all $j \in \mathbb{Z}$. If we let

$$\mathrm{DCT}(f)(k) = \frac{1}{\sqrt{n}} c(k) \sum_{0 \le j < n} f(j) \cos \frac{\pi k(2j+1)}{2n} \text{ for } 0 \le k < n,$$

$$\mathrm{IDCT}(f)(j) = \frac{1}{\sqrt{n}} \sum_{0 \le k < n} c(k) f(k) \cos \frac{\pi k(2j+1)}{2n} \text{ for } 0 \le j < n,$$

where $c(k) = 1$ if $k = 0$ and $c(k) = \sqrt{2}$ otherwise, then DCT and IDCT are inverse operators mapping real-valued signals of finite duration n to signals of the same

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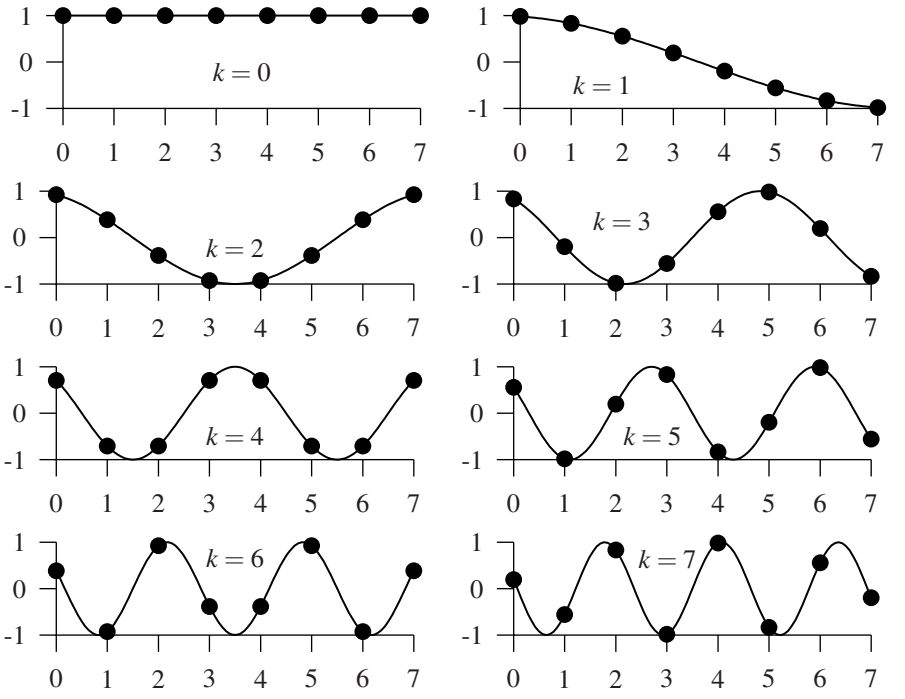


FIGURE 13.3: The discrete cosine signals γ_k for $0 \leq k < 8$.

kind (Exercise 13.6). $\mathrm{DCT}(f)$ is the **Discrete Cosine Transform (DCT)** of f. Exercise 13.6 also shows that computing this transform or its inverse can be reduced to computing a Discrete Fourier Transform, which can be done efficiently via the FFT if n is a power of 2, using $O(n \log n)$ operations in $\mathbb{R}$.

The inversion formula

$$f(j) = (\mathrm{IDCT} \circ \mathrm{DCT})(f)(j) = \frac{1}{\sqrt{n}} \sum_{0 \leq k < n} c(k) \mathrm{DCT}(f)(k) \cos \frac{\pi k(2j+1)}{2n}$$

for $0 \leq j < n$ shows that the Discrete Cosine Transform leads to a representation of the original signal f as a linear combination of periodic signals γ_k, where $\gamma_k(j) = \cos(\pi k(2j+1)/2n)$, with coefficients $c(k)\mathrm{DCT}(f)(k)/\sqrt{n}$. Figure 13.3 depicts the signals γ_k on the interval $0, \ldots, n-1$ for $n = 8$. Larger values of k correspond to a more rapid variation of γ_k.

A possible image data compression algorithm works as follows. For each row $f : \{0, \ldots, n-1\} \longrightarrow \mathbb{R}$ of the image, where $f(j)$ is the luminance of the jth pixel in that row, compute $\mathrm{DCT}(f)(k)$ for $0 \leq k < n$ (if the image is a color image, then, for example, apply this separately for the intensities of each of the three colors red, green, and blue). Then choose a quantizing parameter $q \in \mathbb{R}_{\geq 1}$, divide all values of $\mathrm{DCT}(f)$ by q, and round to the nearest integer. The effect of quantization

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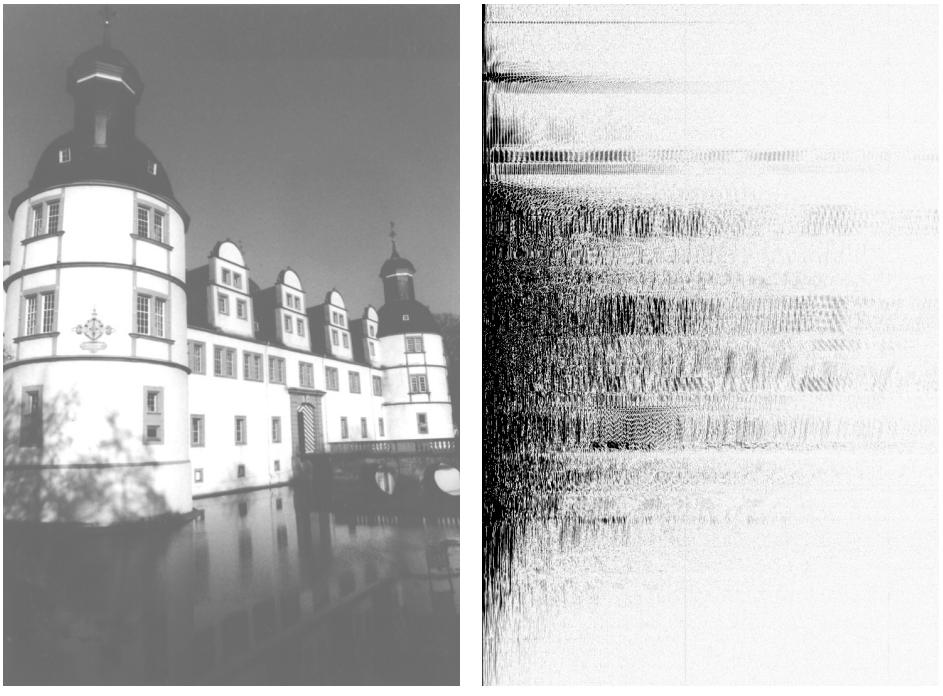


FIGURE 13.4: A grayscale image of Schloß Neuhaus at Paderborn, and the absolute values of its row-wise Discrete Cosine Transform. The color white corresponds to zero, and frequency increases from left to right.

is that those values of $\mathrm{DCT}(f)(k)$ that are close to zero in absolute value (which in general will be the case for the high frequency parts, that is, for large k) vanish completely. Thus large values of q correspond to high compression rates but also to worse image quality. Finally a combination of lossless data compression techniques, such as run length encoding and **Huffman encoding**, is applied to the quantized values. **Run length encoding** compresses each consecutive sequence (= run) of zeroes to two integers: a zero marking the position of the run, followed by the length of the run. For example, the sequence

$$1,2,3,0,0,0,0,4,0,5,-6,0,0,0,1,2$$

is compressed to

$$1,2,3,0,4,4,0,1,5,-6,0,3,1,2$$

thereby decreasing the length by two. We note that "runs" of zeroes of length 1 actually increase the size, like the single zero between 4 and 5 in the above example. In order to reconstruct the image, we proceed in the opposite direction: after decoding the compressed values, we multiply them by q and apply the Discrete Cosine Transform row by row.

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Figure 13.4 shows at left a grayscale image with 1088 rows, 728 columns, and luminance values between 0 (= black) and 255 (= white). Hence the size of the image is $1088 \cdot 728 = 792\,064$ bytes in a dense encoding, where the luminance value of each pixel is stored in one byte. At right, we see the absolute values of the row-wise Discrete Cosine Transforms of that image. The luminance of the kth pixel from the left in row f_l corresponds to the absolute value of $\mathrm{DCT}(f_l)(k)$ (for better visibility, white represents 0 in this image and all values are multiplied by 10), and one can observe that the DCT coefficients get smaller as the frequency increases. Finally, Figure 13.5 shows the same image after quantizing and dequantizing with parameter q (that is, rounding the DCT coefficients to integral multiples of q) and row-wise application of IDCT, for $q = 10$ and $q = 100$. The image at right in Figure 13.4 illustrates why these compression methods are successful. Light gray areas are "rounded down" to 0, and the darker values are similarly simplified. The larger q is, the more rounding occurs.



FIGURE 13.5: The image from Figure 13.4 after quantizing with $q = 10$ and $q = 100$, using a row-wise Discrete Cosine Transform.

Table 13.6 lists the compression rates—the size ratio of the compressed and the original image—for the image from Figure 13.4 with various quantizing parameters q and different lossless data compression techniques. For example, the size of the file after quantizing with $q = 10$ and using both run length and Huffman encoding is 107 646 bytes, which is about 13.59% of the original size, and this

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q	1	2	5	10	20	50	100
Huffman	55.89	44.00	29.37	22.60	18.24	14.89	13.62
Run length	91.81	75.01	43.26	28.52	18.49	8.99	4.64
Run length + Huffman	52.83	39.89	22.26	13.59	8.21	3.85	1.94

TABLE 13.6: Compression rates in % for the image from Figure 13.4 with the row-wise Discrete Cosine Transform and different encoding schemes.

q	1	2	5	10	20	50	100
Huffman	40.95	29.62	20.38	17.13	15.43	14.18	13.52
Run length	78.14	54.35	23.39	12.57	8.08	5.34	4.21
Run length + Huffman	35.14	23.98	11.24	6.14	3.68	2.07	1.33

TABLE 13.7: Compression rates in % for the image from Figure 13.4 with the Discrete Cosine Transform for 8×8 squares and different encoding schemes.



FIGURE 13.8: The image from Figure 13.4 after quantizing with $q = 10$ and $q = 100$, using the Discrete Cosine Transform for 8×8 squares.

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compression rate is the entry in the last row of the column with head 10 in Table 13.6. For comparison, Huffman encoding applied to the image itself (instead of its Discrete Cosine Transform) yields a compression rate of 75.62%, and the GIF graphics format achieves a lossless compression to 54.08% of the original size.

The method described above has the disadvantage that quantization leads to perturbations in the whole row, as can be seen in Figure 13.5 for $q = 100$. Thus the *local* structure of an image (for example, slow variation of the luminance in the sky parts of Figure 13.4) cannot be exploited. This can be circumvented by dividing the image into smaller parts of a fixed size (instead of complete rows) and applying the above compression technique to each part separately.

In the JPEG still image compression standard, for example, the original image is divided into squares of 8×8 pixels, and the two-dimensional Discrete Cosine Transform of each square (which is a combination of a row-wise and a column-wise one-dimensional Discrete Cosine Transform) is computed. Then the DCT coefficients of all squares are quantized, run length encoded and finally Huffman encoded. While the one-dimensional row-wise Discrete Cosine Transform takes only horizontal dependencies into account, its two-dimensional variant covers horizontal and vertical dependencies simultaneously. This together with the improved adaptivity to the local structure of an image leads to significantly higher compression rates than the above row-wise approach (Table 13.7).

Figure 13.8 shows the image from Figure 13.4 after compression and decompression with the Discrete Cosine Transform for 8×8 squares and quantization factors $q = 10$ and $q = 100$. For $q = 10$, for example, one hardly notices any differences between the images in Figures 13.5 and 13.8, but the compression rate of the former is about 13.59%, while the latter compresses down to 6.14%.

Notes. 13.1. Good references for (digital) signal processing are Oppenheim & Schafer (1975) and Oppenheim, Willsky & Young (1983).

13.2. For a description of the Huffman code, see Huffman (1952), §16.3 in Cormen, Leiserson, Rivest & Stein (2009), and Exercise 10.6. The JPEG standard is described in Wallace (1991) and Pennebaker & Mitchell (1993).

Exercises.

13.1 Show that for any discrete periodic signal $f: \mathbb{Z} \longrightarrow \mathbb{C}$ there is a least period (called the **fundamental period**) $n \in \mathbb{N}_{>0}$ such that any other period of f is an integral multiple of n.

13.2 Let $f, g: \mathbb{R} \longrightarrow \mathbb{C}$ be two 2π-periodic signals. If f and g are sufficiently smooth, then the **convolution**

$$(f * g)(t) = \int_0^{2\pi} f(s)g(t-s)ds$$

exists for all $t \in R$. Prove that $f * g$ is again 2π-periodic, and that the convolution property $\widehat{f * g} = \widehat{f} \cdot \widehat{g}$ holds, so that

$$(\widehat{f * g})(k) = \widehat{f}(k) \cdot \widehat{g}(k) \text{ for all } k \in \mathbb{Z}.$$

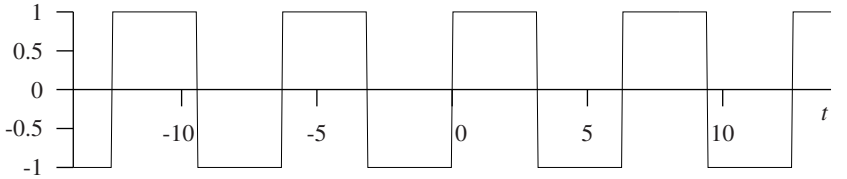
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(Thus the Fourier Transform converts convolution into pointwise multiplication.) You may assume that all occurring integrals exist.

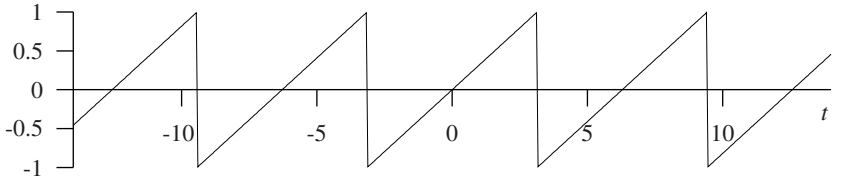
13.3 Let $f(t) = \sin(t) + \sin(10t)/10$. Compute $\widehat{f}(k)$ for $k \in \mathbb{Z}$.

13.4$\longrightarrow$ (i) Compute the Fourier coefficients of $f(t) = e^{int}$ for a fixed $n \in \mathbb{Z}$, where $i = \sqrt{-1}$.

(ii) Compute the Fourier coefficients of the 2π-periodic square wave which has $f(t) = -1$ for $-\pi \le t < 0$ and $f(t) = 1$ for $0 \le t < \pi$.



(iii) Compute the Fourier coefficients of the 2π-periodic triangular wave which has $f(t) = t/\pi$ for $-\pi \le t < \pi$.



13.5 Let $f: \mathbb{Z} \longrightarrow \mathbb{C}$ be a discrete signal of period $n \in \mathbb{N}_{>0}$. Show that $\Re\widehat{f}(k) = 0$ for all k if f is odd, so that $f(j) = -f(-j)$ for all j, and that $\Im\widehat{f}(k) = 0$ for all k if f is even, so that $f(j) = f(-j)$ for all j.

13.6* Let $f: \{0, \ldots, n-1\} \longrightarrow \mathbb{R}$ be a discrete signal of finite duration n. We associate to f a signal $g: \mathbb{Z} \longrightarrow \mathbb{R}$ of period $4n$, by letting

$$\begin{array}{rcl} g(2j+1) & = & g(4n-2j-1) = f(j) \text{ for } 0 \le j < n, \\ g(2j) & = & 0 \text{ for } 0 \le j < 2n, \end{array} \tag{3}$$

and periodically extending g to a function that is defined for all integers. This corresponds to gluing f and a reflected copy of f together and interleaving the result with zeroes. Obviously g is even and $g(j)$ vanishes if j is even.

(i) Prove that the Discrete Fourier Transform $\widehat{g}$ is real-valued of period $4n$ and has the symmetry properties

$$\widehat{g}(k) = \widehat{g}(4n-k) = -\widehat{g}(2n+k) = -\widehat{g}(2n-k) \text{ for } k \in \mathbb{Z}.$$

(ii) Show that the inversion formula

$$f(j) = g(2j+1) = \frac{1}{n}\left(\frac{\widehat{g}(0)}{2} + \sum_{1 \le k < n} \widehat{g}(k) \cos\frac{\pi k(2j+1)}{2n}\right)$$

holds for $0 \le j < n$.

(iii) Conclude that DCT and IDCT are inverse operators.

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